CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

# On the 3x3x3-Lemma

**Tim Van der Linden**

*UCLouvain, Belgium*

19 January 2021 (Tuesday), 16:00 (Lisbon time) · Zoom

**Abstract**

The idea of a short exact sequence of short exact sequences naturally leads to the 3x3-Lemma. What happens when this process is repeated? The aim of this talk is to explain the connection between the concept of a 3-fold extension, short exact sequences of 3x3-diagrams, and a congruence distributivity condition which in the context of an exact Mal'tsev category leads to a denormalised 3x3x3-Lemma. In the context of a semi-abelian category, we ask the question, when a triple of subobjects of a given object induces a 3x3x3-diagram: the answer is non-trivial even in the abelian case. (Joint work with Cyrille Sandry Simeu)